Topic 38 — Pollution (Advanced)

Covers syllabus (click to expand)

Air Pollution (Sources | Air Quality | AQI | NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) | Smog | Photochemical Smog | Industrial | Vehicular | Indoor) | Water Pollution (Surface | Groundwater | Drinking | Standards) | Soil Pollution (Fertilizers | Pesticides | Heavy Metals) | Noise Pollution (Sources | Standards) | Marine Pollution (Coastal | Ocean) | Plastic Pollution (PWM)

Sources baked in: CPCB NAAQS/noise; NCAP 2019; SAFAR; IS 10500; CGWB; PWM Rules; MARPOL; UPPCS PYQs **Weight:** ★★★ High — lichens; smog secondary; BOD; CGWB≠CPCB; Moradabad noise; fly ash **Last verified:** August 2026

Current Affairs (this topic)

- Advanced pollution CA mirrors Pollution & Waste — AQI / CAQM / waste-rule updates.
- Keep criteria-pollutant chemistry pairs consistent with the core pollution chapter.

38.1 Sources of Air Pollution

Identity: Point / non-point | Sectoral | Primary pollutants

- **Point sources** are chimneys, stacks, and pipes; **non-point sources** include roads, farms, and open burning.
- Major sectors: power, industry, vehicles, construction dust, biomass/stubble burning, and households.
- Primary pollutants include **SO_2, NOx, CO, PM, and VOCs**.
- UP winter peaks link stubble smoke, vehicles, industry, and Indo-Gangetic inversion.

Logic: Stubble smoke is a classic non-point seasonal source in north India.

38.2 Air Quality

Identity: NAAQS | Monitoring | Lichens | SAFAR

- **Air quality** is judged against CPCB **National Ambient Air Quality Standards (NAAQS)**.
- Key pollutants monitored: PM2.5, PM10, SO_2, NO_2, O_3, CO, NH_3, Pb, and related species.
- **Lichens** are sensitive bio-indicators of air pollution, especially SO_2 stress.
- They indicate pollution because they are **sensitive** — they do **not** grow well in polluted air.

- **SAFAR** (System of Air Quality and Weather Forecasting and Research) is run with **IITM** support for forecast products.

PYQ — UPPCS Prelims 2025, Q61

1. Lichens are very good indicators of pollution.
2. Lichens grow well in polluted areas.

Options: A. Only 2 B. Neither C. Both D. Only 1

Show answer

Ans: D — Stmt 1 true; stmt 2 false.

PYQ — UPPCS Prelims 2018, Q84

SAFAR inaugurated (July 2018) by:

Show answer

Dr. Harsh Vardhan (option C in that paper).

> **Logic:** Indicator ≠ thrives in dirty air.

38.3 Air Quality Index (AQI)

Identity: CPCB 0–500 | Eight pollutants | Health categories | Sub-indices

- India's **AQI** (CPCB) runs from **0 to 500**. **National Air Quality Index (NAQI)** launched **17 October 2014** under the "one number–one colour–one description" idea.
- Categories: **0–50 Good**, **51–100 Satisfactory**, **101–200 Moderate (नरम दल)**, **201–300 Poor**, **301–400 Very Poor**, **401–500 Severe**.
- Eight pollutants: **PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb**. **CO_2 and CH_4 are not AQI parameters** (they are climate/GHG metrics).
- CO and O_3 use **8-hour** averages; most others use **24-hour** averages. CO is reported in **mg/m³**; NO_2/O_3 typically in **µg/m³**.
- AQI uses sub-indices for pollutants and reports the worst as the day's AQI.
- PM2.5 and PM10 often drive winter AQI in UP NCR/plain cities.
- WHO guideline teaching (updated): annual PM2.5 **≤5 µg/m³**, 24-h PM2.5 **≤15 µg/m³**. **PM2.5** (not PM10) can enter the bloodstream; ozone can trigger asthma. Highest ozone is **not** typically in "inclement weather."

AQI	Category
0–50	Good
51–100	Satisfactory

AQI	Category
101–200	Moderate
201–300	Poor
301–400	Very Poor
401–500	Severe

> **Logic:** Higher AQI = worse air. $CO_2/CH_4 \neq$ AQI.

38.4 National Clean Air Programme (NCAP)

Identity: 2019 | 122 cities | 20–30% PM reduction target

- **NCAP** (राष्ट्रीय स्वच्छ वायु कार्यक्रम) launched in **2019** to improve air quality in non-attainment cities.
- Covers **122 cities** with city action plans, monitoring, and source control.
- Target framework: about **20–30% reduction in PM concentrations** over the programme timeline (baseline-linked).
- UP NCAP cities include Lucknow (लखनऊ), Kanpur, Ghaziabad, Agra, Varanasi (वाराणसी), Meerut (मेरठ), Noida, Prayagraj (प्रयागराज), Moradabad, Bareilly, and others notified over time.
- Complements BS-VI fuel, dust control, and industrial CETP/stack measures.

> **Logic:** NCAP = city clean-air programme, not a groundwater body.

38.5 Smog

Identity: Smoke + fog origin | Secondary pollutant | London type

- **Smog** is a secondary air pollution mixture; classical **London smog** is linked to **SO_2 + smoke** in cold, humid air.
- Smog is a **secondary pollutant** (formed in the atmosphere), unlike CO or fly ash emitted directly.
- wording often links smog to **oxides of nitrogen and sulphur**.

PYQ — UPPCS Prelims 2018, Q121

Which is the secondary pollutant?

Options: A. Smog B. CO_2 C. CO D. Fly ash

Show answer

Ans: A

PYQ — UPPCS Prelims 2019, Q136

Smog is essentially caused by the atmospheric presence of

Options: A. Oxygen and ozone B. Ozone and nitrogen C. Oxygen and nitrogen D. Oxide of nitrogen and sulphur

Show answer

Ans: D

> **Logic:** Smog = secondary. Fly ash = primary (and from thermal plants).

38.6 Photochemical Smog

Identity: NOx + VOC + sunlight → O_3 / PAN | LA type | Eye irritation

- **Photochemical smog** forms when **NOx and VOCs/hydrocarbons** react in strong sunlight.
- Engine path: $N_2 + O_2 \rightarrow NO$; then $NO \rightarrow NO_2$; NO_2 + light → NO + O; $O + O_2 \rightarrow$ **O_3**; hydrocarbons + NOx → **PAN**.
- It always involves **ozone** as a key secondary product; **PAN**, formaldehyde, and aldehydes also appear.
- **PAN (peroxyacetyl nitrate)** is a powerful **eye irritant** (lachrymator) and damages plant chloroplasts.
- Typical of traffic-rich sunny cities (Los Angeles type); also relevant to Indian summer/pre-monsoon urban (नगरीय) episodes.
- The brown-orange haze of photochemical smog is a **secondary** pollution state — SO_2 stack gas alone is **primary**.

PYQ — UPPCS Prelims 2022, Q46

Photochemical smog always contains—

Options: A. Phosphorus B. Ozone C. Methane D. Aluminium ion

Show answer

Ans: B

> **Logic:** Photochemical smog ≠ classical London SO_2 fog. PAN ≠ primary stack emission.

38.7 Industrial Emissions

Identity: Stacks / SPM / SO_2 | Fly ash | CETP | Kanpur tanneries

- Industry emits PM, SO_2, NOx, VOCs, and process-specific toxins.
- **Fly ash** pollution is caused by **thermal power plants**.
- Fly ash is used for **bricks** and as a **Portland cement** replacement in concrete.
- Main chemical components: **SiO_2, Al_2O_3**, and sometimes **CaO**, plus toxic elements such as **lead, arsenic, and copper** — not "oxides only."
- Control tools: ESPs, **bag filters** (finer PM, often **<~50 µm**), **cyclone separators/collectors** (coarser, **>~50 µm**), FGD, wet scrubbers, and **CETPs** for clusters.
- **Cyclone divider** is **not** a standard instrumental device name for air-pollution control.
- Mercury teaching: **gold mining** and **coal TPPs** are major sources; **no known safe level** of mercury exposure.
- UP geography: Kanpur tanneries (chromium/organics), Singrauli thermal belt, brick kilns.

PYQ — UPPCS Prelims 2021, Q143

Flyash pollution is caused by

Options: A. Oil refining B. Fertilizer industry C. Thermal power plant D. Mining

Show answer

Ans: C

> **Logic:** Fly ash = thermal power **primary** pollutant; smog = secondary. Statement "fly ash = SiO_2 and CaO only, no toxics" is false.

38.8 Vehicular Emissions

Identity: CO / NOx / PM / VOCs | BS-VI 2020 | Urban traffic

- Vehicles emit **CO, NOx, PM, VOCs**, and CO_2; incomplete combustion and cigarette smoke classically yield **CO**.
- Lead antiknock in petrol was a major source of airborne lead toxicity.
- India moved to **BS-VI** norms nationwide from **1 April 2020**, leapfrogging BS-V after **BS-IV** from **1 April 2017**.
- Euro norms limit CO, hydrocarbons, NOx, and PM. For **Euro-II**, ultra-low sulphur diesel needs sulphur **≤0.05%**.
- **Catalytic converters** use **platinum, palladium, and rhodium** to convert CO → CO_2, NOx → N_2, and HC → CO_2 + H_2O.
- Congestion, diesel fleets, and two-wheelers drive city NO_2/PM peaks.
- Complementary tools: EV push, PUC checks, odd-even-type episodic measures, CAFE efficiency norms.
- **H-CNG** (hydrogen-enriched CNG) can cut CO/HC/CO_2 versus CNG; hydrogen up to about **one-fifth** by volume can be blended; it does **not eliminate** CO and is **not** cheaper than CNG.

- **CCAC** (Climate and Clean Air Coalition (गठबंधन), **2012**, with UNEP) focuses on **methane, black carbon, and HFCs** — statement that it is a unique G20-only initiative is false.

> **Logic:** BS-VI cut sulphur in fuel and tightened NOx/PM limits versus BS-IV. CCAC ≠ G20 club.

38.9 Indoor Air Pollution

Identity: Biomass chulha | PM2.5 | Women/children exposure

- Indoor pollution comes from **biomass cookstoves**, kerosene, incense, and tobacco smoke.
- Cigarette smoke contains **CO, HCN, NOx, formaldehyde, acrolein, benzene**, and small amounts of **nicotine**.
- Benzene exposure sources include automobile exhaust, tobacco smoke, wood burning, varnished furniture, and polyurethane products.
- The most significant classic **indoor** pollutant gas in many stems is **radon** — colourless, odourless, radioactive, from soil/radium, linked to lung cancer.
- Rural indoor PM2.5 can exceed outdoor levels and heavily affects women and children.
- Clean cooking (LPG/PNG/electric) is the main mitigation path (Ujjwala-type coverage).
- **Asian Brown Cloud** / atmospheric brown cloud is especially noted over **South Asia** (often Jan–Mar), rich in aerosols including black carbon and sulphate.

> **Logic:** Indoor air is a household health issue, not only an outdoor AQI story. Radon ≠ SO_2 as "most important indoor."

38.10 Surface Water Pollution

Identity: BOD / COD / DO | Sewage / industry | Eutrophication

- Surface waters are polluted by sewage, industrial effluent, and agricultural runoff.
- **BOD** measures oxygen used by microbes decomposing organic matter — a standard **aquatic pollution assay**.
- Higher BOD means worse organic pollution; **COD ≥ BOD**.
- Clean-ish river fact: **DO > ~6 mg/L**, **BOD < ~3 mg/L**. Teaching banks often call DO **<4 mg/L** highly polluted.
- **Eutrophication:** excess **N and P** → algal bloom → DO crash → fish kill. Eutrophication is **not** an air-pollution process.
- Highest Ganga **BOD** stretch is classically **Kanpur–Allahabad**. **Damodar** is the classic **biological desert** river.
- Leather-industry effluent is a top chemical pollution source for water and soil.
- UP rivers: Ganga (Kanpur stretch), Gomti (Lucknow), Yamuna (Mathura (मथुरा)–Agra influence).

PYQ — UPPCS Prelims 2023, Q103

Biological Oxygen Demand (BOD) (जैव रासायनिक ऑक्सीजन मांग) is a standard criterion for—

Options: A. Pollution assay in aquatic ecosystems B. High-altitude O_2 C. Blood O_2 D. Forest O_2

Show answer

Ans: A

> **Logic:** BOD = water organic pollution test, not blood oximetry.

38.11 Groundwater Pollution

Identity: Arsenic / fluoride / nitrate | CGWB reports

- Groundwater contaminants include **arsenic, fluoride, nitrate, iron, salinity**, uranium, and industrial leachates.
- **Arsenic** contamination is classically associated with the **Bangladesh** / lower Gangetic basin story; first Indian notice often dated to **West Bengal, 1978**.
- Affected Indian geography teaching: West Bengal, Jharkhand, Bihar, UP, Assam, Manipur, and parts of Chhattisgarh.
- Chronic arsenic exposure links to cancers and **Black Foot** disease teaching. WHO guideline often cited as **0.01 mg/L**.
- **Fluoride** excess causes fluorosis and **knock-knee** syndrome. **Nitrate** links to blue-baby risk.
- The **Annual Groundwater Quality Report** is prepared by the **Central Ground Water Board (CGWB)** — not CPCB.

PYQ — UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 prepared by:

1. CPCB 2. CWC (केंद्रीय जल आयोग) 3. CGWB

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

Show answer

Ans: B

PYQ — UPPCS Prelims 2022, Q63

Which country has natural arsenic water contamination?

Options: A. Sri Lanka B. Bangladesh C. Bhutan D. Pakistan

Show answer

Ans: B

> **Logic:** CGWB = groundwater. CPCB = pollution control board standards/monitoring.

38.12 Drinking Water Pollution

Identity: Pathogens | Coliform | Fluorosis / arsenicosis | IS 10500

- Unsafe drinking water carries pathogens (diarrhoea, cholera, typhoid risk) and chemical toxins.
- Microbial safety uses **coliform / E. coli** indicators.
- Chronic chemical risks: **fluorosis** (fluoride), **arsenicosis** (arsenic), nitrate (blue-baby risk in infants).
- Treatment chain: source protection → filtration/disinfection → safe storage.

> **Logic:** Microbial and chemical risks are different drinking-water failure modes.

38.13 Water Quality Standards

Identity: Water Act 1974 | BIS IS 10500 | CPCB river classes

- **Water (Prevention and Control of Pollution) Act, 1974** is the core water-pollution statute.
- **BIS IS 10500** sets drinking-water specification limits.
- CPCB classifies river stretches for designated best uses (A–E style classes).
- CPCB/SPCB enforce consent, standards, and monitoring for effluent and ambient water.

> **Logic:** Drinking standard (IS 10500) ≠ ambient river class alone.

38.14 Fertilizers and Environment (पर्यावरण)

Identity: N/P runoff | Eutrophication | Urea overuse

- Excess **nitrogen and phosphorus** fertilisers run off into water bodies.
- Result: eutrophication, nitrate groundwater rise, and N_2O climate link (Topic 37).
- Indo-Gangetic wheat–rice systems are high-input geography.

> **Logic:** Fertiliser pollution is mostly nutrient overload, not heavy-metal tanning.

38.15 Pesticides and Environment

Identity: Persistence | Biomagnification (जैव आवर्धन) | Not biodegradable

- Many pesticides are **persistent** and can **biomagnify** along food chains.

- UPPCS treats **pesticide** as **not a biodegradable pollutant** versus domestic waste/urine/faecal matter.

PYQ — UPPCS Prelims 2023, Q120

Which is not a biodegradable pollutant?

Options: A. Domestic waste B. Urine C. Faecal matter D. Pesticide

Show answer

Ans: D

> **Logic:** Silent Spring-era logic: persistent pesticides harm ecosystems.

38.16 Heavy Metal Pollution

Identity: Pb / Cd / Cr / Hg | Bioaccumulation (जैव संचय) | Kanpur Cr

- Heavy metals do not degrade; they **bioaccumulate**.
- Key facts: **Pb** (batteries/paint/antiknock petrol legacy), **Cd** (batteries/industry — **Itai-itai**), **Cr** (tanneries — Kanpur), **Hg** (**Minamata**; gold mining + coal TPP sources; **no known safe exposure level**).
- Asbestos dust → lung disease / mesothelioma teaching. Soil particles → breathing ailments. NOx → cancer risk teaching in some match lists. CO displaces oxygen in blood.
- Soil and sediment act as long-term sinks near industrial clusters.

Metal / pollutant	Classic disease / effect
Cr	Kanpur tanneries
Hg	Minamata / neurological
Pb	Neurotoxin; IQ / CNS
Cd	Itai-itai / bone-kidney
As	Black Foot / cancers
F	Knock-knee / fluorosis
Asbestos	Lung scarring / cancer

> **Logic:** Heavy metals ≠ biodegradable organics. Fluoride ≠ “indigestion only.”

38.17 Sources of Noise Pollution

Identity: Traffic / industry / construction | Moradabad UNEP fact

- Main sources: road traffic, industry, construction, railways, aircraft, loudspeakers, and generators.
- Jet flights are among the most intense common outdoor noise sources; sonic boom = Mach-scale teaching.
- **Green muffler** = multi-row roadside tree belts that absorb noise (and dust).
- **Moradabad** was listed as the world's **second most noise-polluted city** in UNEP **Frontier Report 2022**.
- **Ringelmann scale** (रिंगेलमैन पैमाना) measures smoke density, not noise.

PYQ — UPPCS Prelims 2022, Q81

UNEP Frontier Report 2022 — second most noise-polluted city:

Options: A. Kota B. Moradabad C. Indore D. Patna (पटना)

Show answer

Ans: B

> **Logic:** Moradabad = noise PYQ city, not only brassware geography.

38.18 Noise Standards

Identity: CPCB zone limits | dB scale | Day/night

- Noise is measured in **decibels (dB)** on a logarithmic scale.
- CPCB ambient daytime limits (standard fact): **Silence 50**, **Residential 55**, **Commercial 65**, **Industrial 75** dB; night limits are lower.
- Silence zones include areas around hospitals, courts, and educational institutions.

PYQ — UPPCS Prelims 2022, Q115

Match area → permissible level (paper List-II used 50/55/65/70):

Show answer

Intended Ans: C — Residential **55**, Silent **50**, Industrial **70** (as listed), Commercial **65**. Teach CPCB industrial day standard as **75 dB** in general notes.

> **Logic:** Silence < Residential < Commercial < Industrial.

38.19 Coastal Pollution

Identity: Land runoff | Ports / sewage | Mangrove stress

- Coastal pollution comes mainly from **land-based** sewage, industry, ports, aquaculture, and plastics.
- Damages mangroves, estuaries, turtle nesting beaches, and nearshore fisheries.
- India's first **National Centre for Marine Biodiversity (NCMB)** is at **Jamnagar (Gujarat)** — coastal biodiversity safeguarding, not a CPCB/NGT fact.

> **Logic:** Coastal pollution is mostly land-origin, not only ship dumps.

38.20 Ocean Pollution

Identity: MARPOL | Oil / plastic / nutrients | ~80% land-based

- About **80%** of marine pollution originates on **land**.
- **MARPOL** is the key international ship-pollution convention. **IMO** headquarters is in **London**.
- **Ocean acidification** (CO_2 → falling ocean pH) harms calcareous phytoplankton, coral reefs, and some animals with phytoplankton larvae — cloud seeding is **not** the standard concern list.
- Harmful algal blooms link to estuary (ज्वारनदमुख) nutrients, monsoon run-off, and upwelling.
- **SACEP** = South Asia Cooperative Environment Programme.
- Major issues: oil spills, nutrient dead zones, and plastic/microplastic accumulation in gyres.
- Coral stress combines warming and local pollution (cross-topic with climate/coral bleaching (प्रवाल विरंजन)).

> **Logic:** Ocean pollution ≠ only mid-ocean dumping. Acidification ≠ cloud seeding.

38.21 Plastic Waste Management

Identity: PWM Rules 2016 | SUP ban 2022 | EPR | Microplastics

- **Plastic Waste Management Rules, 2016** (amended later) govern collection, recycling, and producer duties.
- **Single-Use Plastic (SUP)** ban notification effective **July 2022** for identified items.
- **EPR** makes producers responsible for end-of-life plastic; EPR featured early in plastic/e-waste rules from **2011**.
- India is among the world's largest plastic-waste generators (~**3rd** in common framing).
- **Microplastics** are plastic particles **< 5 mm**. **Microbeads** in cosmetics are harmful to **marine ecosystems**.
- Polythene bags persist because they are **polymers** of ethylene, not because they are proteins.

> **Logic:** PWM/EPR = management law track; SUP ban = item-specific prohibition (प्रतिषेध) track.
